COGS 9 Section

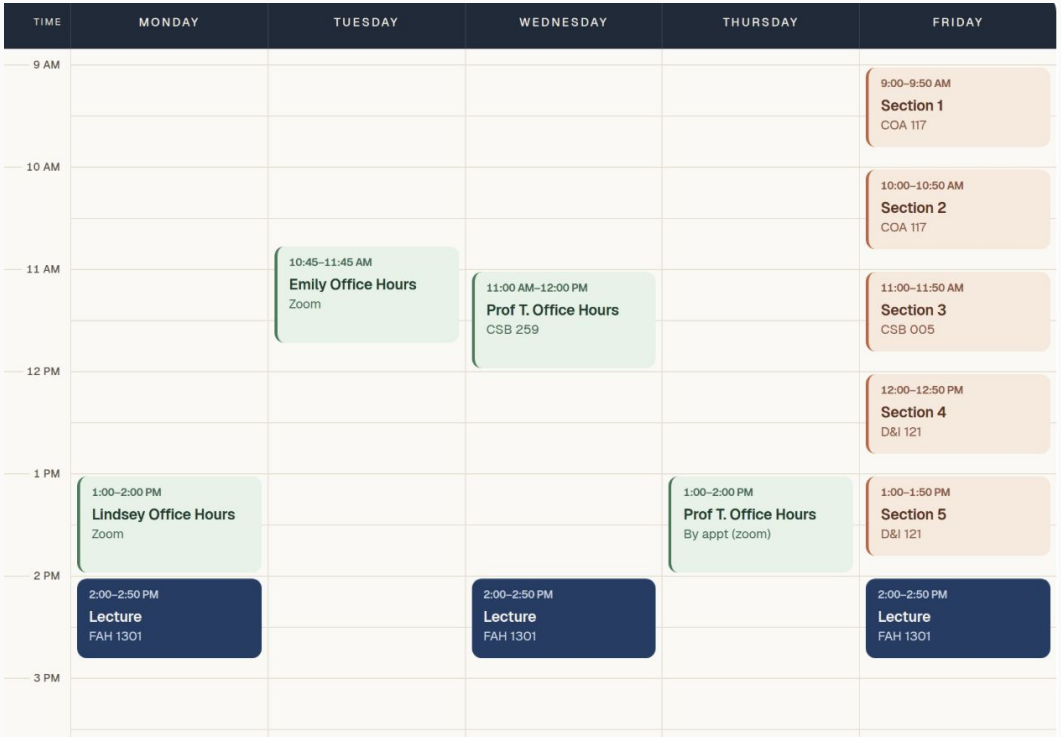
TA: Lindsey Gu

Friday, October 9

Office Hours

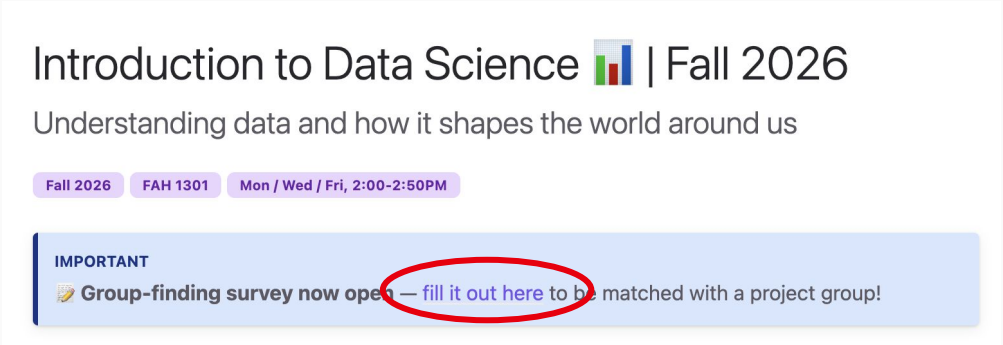
TA office hours are **Mondays/Tuesdays**

- Posting questions: **Piazza**
- Please send an email for further questions, no Canvas messages!



Reminders

- Due Wed @ 11:59 (can all be found on course website)
  - Reading Quiz 2
  - Group-finding survey!
    - Must fill it out if you want to be matched to a group
    - If you miss the deadline, you have to find your own group (5 person limit)
- If you are joining late, you have until Friday of Week 3 to submit missed assignments!
  - Pre-course survey
  - Syllabus quiz
  - Reading quiz 1
  - Reading quiz 2



Questions?

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Are you ready for Week 3?

A. 😀 B. 😎 C. 🤔 D. 😅 E. 😴

Real-World Data Ethics Cases

Each of the following incidents illustrates how ethical failures can occur at different stages of the data science process:

- Facebook–Cambridge Analytica (2014–2018): Tens of millions of users' data were harvested without consent and used for targeted political advertising — **an example of harmful data collection.**
- Strava Heatmap (2018): Aggregated GPS exercise data from soldiers inadvertently exposed the locations of secret military bases — **a failure of data sharing review.**
- Amazon Hiring Algorithm (2018): A machine learning model trained on historically male-dominated resumes learned to discriminate against women — **a data modeling problem caused by class imbalance.**
- Target Pregnancy Prediction (2012): Purchase history was used to infer and act on sensitive personal information (pregnancy) without the individual's awareness — **raising data analysis ethics questions.**

Key Takeaway: Ethical risks exist at every stage — collection, sharing, analysis, and modeling. Responsible data scientists ask "Should we?" not just "Can we?"

Data Types & Matching Them to Visualizations

Type	Examples	Best Visualization
Qualitative – Nominal (categories without order)	Eye color, major, blood type	Bar chart, pie chart
Qualitative – Ordinal (with ranked order)	Survey ratings (low/med/high), letter grades	Bar chart, grouped bar chart
Quantitative - Discrete (countable value)	# of car owned, # of students in a classroom	Bar chart, histogram,dot plot
Quantitative - Continuous (any value in range)	Height, weight, temperature	Histogram, line chart, scatter plot, box plot

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Visualization

Two students are given the same dataset showing monthly sales across four product categories over three years.

- Student A creates a grouped bar chart for each month
- Student B creates a line chart with one line per category

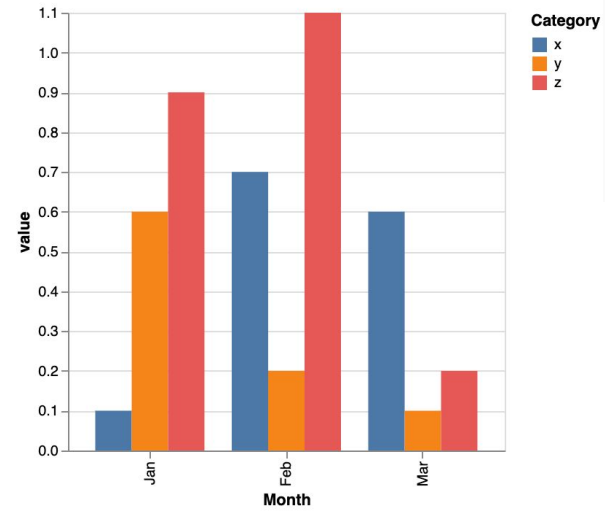
They are asked: “Which categories are increasing or decreasing over time?”

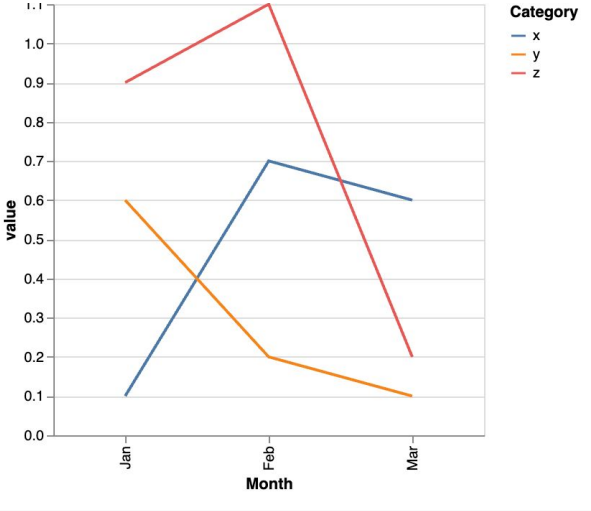
Which explains which visualization is more appropriate and why?

(A) Student A’s grouped bar chart because it allows precise comparison of monthly values

(B) Student B’s line chart because position along a continuous axis makes trends over time easier to perceive

(C) Both charts are equally effective because they show the same data





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This or That!

A company removes names and emails from their dataset before sharing it publicly. This means the data is safe to release and fully anonymous.

WEBCLICKER:

A. True – removing identifiers makes it safe
B. False – other attributes can still re-identify people
C. Complicated – depends on what other data is out there

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this is the PII fallacy from the Myths & Fallacies reading: Removing names/emails doesn't make data safe if things like location patterns, demographics, or behavior can still fingerprint someone.

Reading 2A “Data’s Day of Reckoning”: What is the problem?

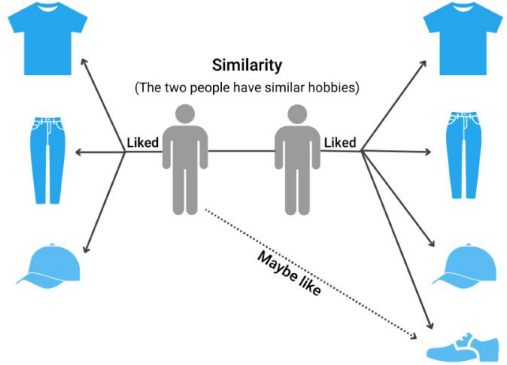
- Data/AI systems can be useful, but they **can also be weaponized**
- **Harm** comes from the fact that these systems are built inside organizations with incentives like:
  - speed
  - growth
  - profit
  - weak accountability
- The authors argue that tech is **not neutral**
- **Main idea:** harm does not just come from “bad people,” but from everyday design decisions and systems that ignore the public good

What does “taking responsibility” actually mean?

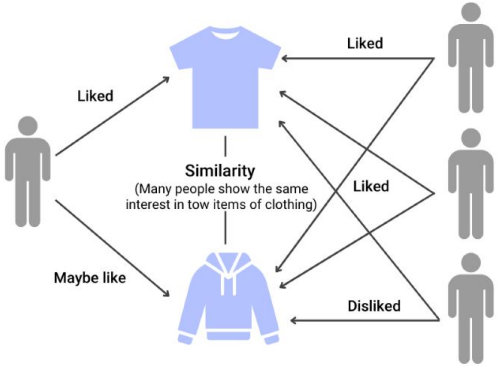
- Ethics and security must be integrated into technical education and product development
- **Responsibility must be made practical**, not abstract
- The reading proposes:
  - ethics/security in the core curriculum
  - **checklists** during design and deployment
  - routines for **consent, bias checks, abuse cases, and monitoring**
  - **ability to pause or shut down** harmful systems
  - workplace cultures where employees can raise concerns safely
  - more diverse teams to catch blind spots

Real-world example and why regulation matters

- **Example: social media recommendation systems**
  - optimized for engagement
  - can amplify misinformation, harassment, or extremism
  - shows how products can cause harm when incentives do not align with the public good



User-based



Item-based



Real-world example and why regulation matters

- **The reading says regulation may become necessary when:**
  - companies will not self-correct
  - harms affect the broader public
  - internal ethics processes are too weak
- **But regulation also has limits:**
  - policy usually moves slower than technology
  - lawmakers may not fully understand technical systems
  - regulation cannot replace ethical design inside organizations

Questions?

Reading 2B Myths & Fallacies of PII *(Personal Identifiable Information)*

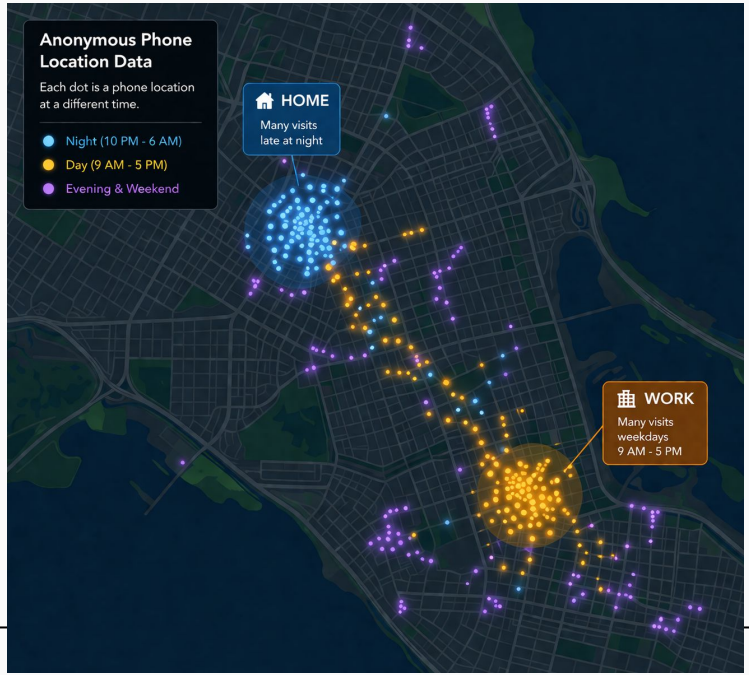
- Many people assume privacy is protected once names or IDs are removed
- The reading calls this the **PII fallacy**
- PII is hard to define because **identification is context-dependent**
- Privacy laws often define it broadly, but technical practice often treats it as a fixed list
- **Main point:** there is no clear, permanent line between:
  - identifying data
  - non-identifying data

How re-identification / linkage attacks work

- **A person can be identified without their name being present**
- Re-identification works by linking “anonymous” data to other datasets using shared patterns
- **This can happen through:**
  - demographics
  - search history
  - location
  - patterns
  - social network structure
- **Key idea:** any sufficiently rich set of attributes can act like a fingerprint
- So-called **anonymous data becomes identifiable** when combined with outside information

Real-World Example: Anonymous Location Data

- A company shares “anonymous” phone location data
- Names and IDs are removed
- But the data still shows:
  - where someone **sleeps**
  - where they **work**
  - where they **go** often
- That pattern can reveal who they are
- **Main idea:** behavior itself can identify a person



Better privacy protection and its tradeoffs

- The reading argues we should move beyond simple de-identification
- Better approaches include:
  - **case-by-case** privacy analysis
  - **interactive/query-based access** instead of releasing raw data
  - stronger **access controls**
  - **informed consent**
  - contracts limiting acceptable data use
  - privacy-preserving methods like **differential privacy**
- **Tradeoffs:**
  - harder and more expensive to manage
  - may reduce convenience or data utility
  - require ongoing oversight, audits, and infrastructure

Questions?

Meet & Greet

1. Have you thought about whether you are comfortable or uncomfortable with companies using your data to increase their sales? What specific data uses feel “fine” vs. “creepy”/violation of your privacy?
2. Think about 3–5 data points you’ve shared recently (location, purchase, fitness, clicks, photos, messages). Which ones feel non-identifying—and how could they still identify you when combined with other data?
3. What are you getting in exchange for data collection (discounts, convenience, personalization, social connection)? Are those benefits worth giving your data for?

Form a group of 3-4 people, introduce yourselves and choose one of the question to discuss!

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